Day 5: Assemble and Disassemble a PCPractical (1 hour): Full hands-on session: Assemble and disassemble a desktop PC.Task: Document the process with labeled steps.

1.List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

ANS:

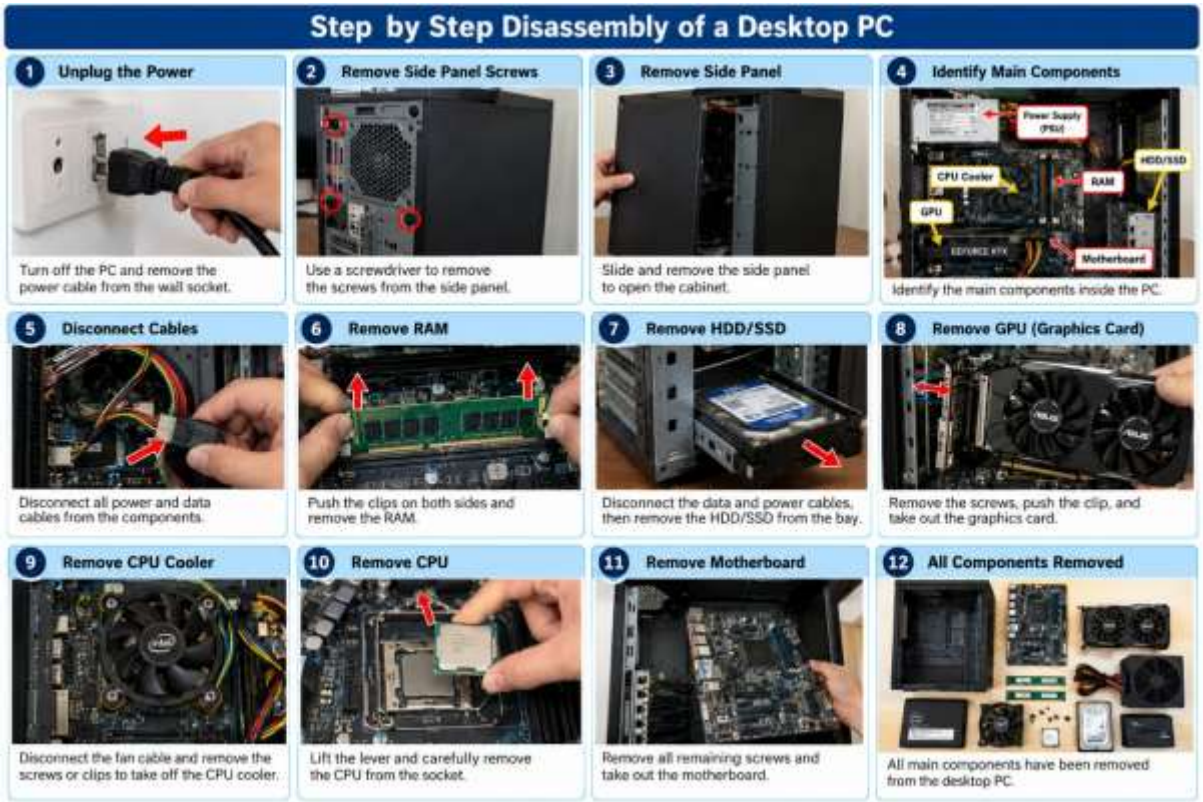
No.	Component	Short Description
1	Motherboard	The main board that connects all computer components.
2	CPU (Processor)	The brain of the computer. It processes instructions and performs tasks.
3	RAM	Temporary memory that helps the computer run programs.
4	HDD	Stores files, photos, videos, and software.
5	SSD	Stores data and helps the computer start and open apps faster.
6	GPU (Graphics Card)	Displays images, videos, and games on the screen.
7	Power Supply (PSU)	Supplies power to all computer components.
8	Cooling Fan	Moves air to keep computer parts cool.
9	Heatsink	Takes heat away from the CPU or other components.
10	CPU Cooler	Keeps the CPU from becoming too hot. It may use a fan and heatsink.
11	Cabinet / Case	Holds and protects all the internal components.
12	Network Card (NIC)	Connects the computer to a network or the internet. It may be built into the motherboard.
13	Sound Card	Handles audio input and output. It is usually built into the motherboard.
14	CMOS Battery	Helps maintain the system clock and BIOS settings when the PC is off.

15	Optical Drive	Reads CDs or DVDs, if the computer has one.
16	SATA Cables	Connect SATA storage drives to the motherboard.
17	Power Cables	Carry electricity from the power supply to components.
18	Expansion Cards	Add extra features, such as better graphics, sound, or networking.

**2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

ANS:



Step 1: Turn Off the PC Shut down the computer properly.

Step 2: Unplug the Power Cable Remove the power cable from the wall socket.

Step 3: Remove the Side Panel Unscrew the cabinet screws and open the side panel.

Step 4: Identify the Components Find the motherboard, CPU, RAM, HDD/SSD, GPU, and power supply.

Step 5: Disconnect the Cables Carefully disconnect the power and data cables.

Step 6: Remove the RAM Open the RAM slot clips and gently remove the RAM.

Step 7: Remove the HDD/SSD Disconnect the cables and remove the storage drive.

Step 8: Remove the GPU Unscrew the graphics card and carefully remove it from the PCIe slot.

Step 9: Remove the CPU Cooler Disconnect the fan cable and carefully remove the cooler.

Step 10: Remove the CPU Open the CPU socket lock and carefully lift out the processor.

Step 11: Remove the Motherboard Unscrew the motherboard and carefully take it out of the cabinet.

Step 12: Arrange All Components Place the removed components safely on a clean surface.

3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

ANS:

Step 1: Install the Motherboard

Place the motherboard inside the cabinet and tighten the screws. It connects all the main components.

Step 2: Install the CPU

Carefully place the CPU in its socket and lock it. The CPU processes computer instructions.

Step 3: Install the CPU Cooler

Attach the cooler and connect its fan cable. It prevents the CPU from overheating.

Step 4: Install the RAM

Insert the RAM into the correct slots until it locks into place. RAM helps programs run smoothly.

Step 5: Install the HDD/SSD

Place the storage drive in its bay and connect the required cables. It stores files and software.

Step 6: Install the GPU

Insert the graphics card into the PCIe slot and secure it with screws. It handles graphics and games.

Step 7: Connect All Cables

Connect the motherboard power cable, CPU power cable, storage cables, and other required cables. These supply power and transfer data.

Step 8: Check All Connections

Check that every component and cable is connected properly. This helps prevent problems when starting the PC.

Step 9: Close the Cabinet

Replace the side panel and tighten the screws. This protects the internal components.

Step 10: Connect the External Devices

Connect the monitor, keyboard, mouse, and power cable.

Step 11: Turn On the PC

Switch on the computer and check whether it starts correctly.

Step 12: Check the System

Open Windows and check that the RAM, storage drive, and graphics card are detected.

The motherboard must be installed securely, and the CPU cooler must be fitted correctly before powering on. Connecting the cables and checking all components before starting helps prevent damage and boot problems.

4.Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

ANS:

1. Tools Needed

Tool	**Why is it important?**
Screwdriver	To remove and tighten screws.
Anti-static wrist strap	Helps protect computer parts from static electricity.
Small container	To keep screws safe.
Soft brush	To remove dust carefully.
Cable ties	To organize cables neatly.
Thermal paste	Helps transfer heat from the CPU to the cooler when needed.

2. Safety Precautions

Safety precaution	**Why is it important?**
Turn off the PC	Prevents electrical problems.

Unplug the power cable	Prevents electric shock and damage.
Keep hands dry	Reduces electrical risk.
Touch a grounded metal surface	Helps reduce static electricity.
Hold components by their edges	Protects delicate parts and contacts.
Do not force components	Prevents damage to slots and connectors.
Keep liquids away	Prevents short circuits.
Take photos before removing cables	Helps reconnect cables correctly.
Keep screws organized	Prevents losing screws.
Follow the manufacturer's manual	Ensures correct installation.

3. Important Reminder

Always disconnect the power before opening the PC.

Never open the power supply unit itself.

Check all connections before turning on the computer.